

International Institute of Information Technology, Hyderabad
(Deemed to be University)

GS0.301: Introduction to Spatial Science and Technology
End Semester Examination

Date: 30/04/2026

Max. Time: 1.5 Hrs. **Max. Marks:** 40

Note:

1. Answer all questions. There are five sections. Section I (4 marks), Section II (4 marks), Section III (12 Marks), Section IV (10 Marks), and Section V (10 Marks).
2. Make technically valid assumptions wherever necessary and state your assumptions clearly.
3. Wherever required provide necessary figures to better explain the answers.

[Section I: Multiple choice questions, 1 Mark each]

1. Taxi GPS trajectories, User's mobile location data, Swiggy's delivery partner's movement data are examples of:
 - a. Static vector data
 - ☒ b. Spatio-temporal data
 - c. Temporal only data
 - d. Network data
2. The attribute road class (NH-National Highway, SH-State Highway, etc) is of type:
 - ☒ a. Nominal
 - b. Cyclical
 - ☒ c. Ordinal
 - d. Interval

either
3. Which visualization is best for showing discrete locations with varying quantity of voter's turn around?
 - a. Choropleth
 - ☒ b. Proportional symbol
 - c. Isoline
 - ☒ d. Heat map

(2nd choice)
4. What does "scale" in a toposheet map refer to?
 - a. Color gradient
 - b. Data format
 - ☒ c. Ratio of map distance to ground distance
 - d. Projection type

[Section II: Answer following questions 1 Mark each]

5. While visualizing fires from space, we see an active fire front (red hotspot) in an False Color Composite (FCC) with SWIR, NIR, Red (RGB) combination and smoke plume in an FCC with NIR, Red, Green (RGB) combination. Provide 1-2 line justification?
6. List various water and climate related disasters.
7. What is the physical basis of retrieving fires from satellite data? What are the optimal wavelength bands used to pick emittance from fires?
8. Polar orbiting vs. geostationary satellites - which one would you recommend for fire monitoring? and why?

[Section III: Answer following questions, 2 Marks each]

9. What are the two primary types of geospatial data? With one example of each, explain the fundamental difference in their spatial representation.
10. Recall the London cholera map prepared by Dr. John snow. Which geospatial visualization technique will be appropriate to represent the cholera outbreak? Explain with the help of a figure.
11. List any four sources of spatial data and give two examples of each.
12. What is the main purpose of map projections and what is the role of Ellipsoid in map projections?
13. Explain with the help of example/figure, why the raster imagery gets pixaleted when you zoom-in?

14. What are the different spatial scales while solving water resources systems management problems. Give two examples of representing a water resources system in terms of hydrological variables?

[Section IV: Answer the following questions, 5 Marks each]

15. You are given a district-level rainfall data for the state of Maharashtra over a monsoon season.
- Which map visualization technique would you choose? Justify your choice. [2 M]
 - How would your choice change if you had point-based rainfall station data instead? (1 M)
 - Discuss your approach to use the point-based rainfall data to obtain rainfall at every other location in the state. [2 M]
16. We don't know the accuracy of the AI model that was used to generate the Dynamic World LULC map. You are tasked to assess the accuracy of DW LULC map of your chosen districts. Answer the following:
- State briefly the method you would use to assess the accuracy of the map. [1 M]
 - You would need validation data for the accuracy assessment. How would you collect or obtain validation data for accuracy assessment? [1 M]
 - Propose and explain any metric to quantify the accuracy. [1 M]
 - Discuss how these inaccuracies will impact your change analysis and how to account for such inaccuracies in the final LULC change analysis? [2 M]

[Section V: Answer the following questions, 10 Marks]

17. Assuming you have access to the repeated coverage of the flood affected area from the satellite data. Think about a conceptual framework that can be used during flood response and recovery phase. Answer the following:
- State in English one query for each phase that can be used during flood response and recovery phase. [2 M]
 - List and justify what dataset would you need (you can go beyond the satellite imagery and be comprehensive in explaining their roles) [3 M]
 - Explain how will you answer the queries mentioned above (in (a)) using spatial and non-spatial operations? [2 M]
 - Draw a conceptual framework that can be used during flood response and recovery phase by any disaster response authority. Clearly show input datasets, operations, outputs, etc. [3 M]